

SAI PRASANNA

PROFILE

 Data & Cloud Engineer with 4 years of experience designing, implementing, and managing secure, scalable cloud and data platforms across enterprise and healthcare environments. Proven expertise in Azure cloud architecture, infrastructure automation, DevOps, and enterprise security. Delivered Azure Health Data Services (AHDS) solutions supporting FHIR R4 interoperability, CMS compliance, healthcare data exchange, and API-driven integration initiatives. Experienced in implementing Patient Access, Provider Directory, Payer-to-Payer, and Prior Authorization solutions aligned with healthcare interoperability standards. Skilled in SMART on FHIR authorization, Azure API Management, healthcare data pipelines, and cloud-native integration patterns. Strong technical background in Infrastructure as Code (Terraform, Bicep), Azure DevOps CI/CD, Azure Kubernetes Service (AKS), FinOps governance, monitoring, and cloud security best practices.

EXPERIENCE

Cloud Engineer | Apr 2026 – Present

-  Designed, implemented, and supported secure, scalable Azure cloud architectures, healthcare data platforms, and enterprise data integration solutions across cloud-native environments. Automated infrastructure provisioning using Terraform and Bicep for Azure resources including VNets, Private Endpoints, Key Vault, Storage Accounts, Azure Functions, Azure API Management, Application Gateway, Azure SQL, Cosmos DB, Azure Monitor, and Log Analytics.
-  Built and supported Azure Health Data Services (AHDS) solutions for FHIR R4 interoperability, CMS compliance, healthcare data exchange, and enterprise API integrations. Developed ETL pipelines to ingest, transform, and validate clinical and claims data into standardized FHIR resources, supporting Patient Access, Provider Directory, and healthcare data-sharing requirements.
- Configured Azure API Management for secure API routing, OAuth2 and SMART on FHIR authorization, policy enforcement, and developer access management. Implemented secure hub-and-spoke networking, private connectivity, DNS integration, RBAC, Managed Identities, Azure Key Vault, Azure Policy, and network security controls to support HIPAA, CMS, and enterprise security standards.
- Developed Azure Workbooks dashboards using Azure Monitor, Log Analytics, and Azure Policy data to provide visibility into platform health, compliance, governance, and operational performance. Implemented FinOps practices including log ingestion optimization, retention management, budget monitoring, and tagging-based cost allocation.
- Integrated Azure DevOps deployment pipelines with ServiceNow CMDB to automate configuration management and improve audit readiness. Supported Azure SQL and Cosmos DB environments including schema design, indexing, performance tuning, and data management. Integrated Azure Sentinel, Logic Apps, and ServiceNow for automated incident management and improved operational response.
- Collaborated with architects, DevOps teams, developers, security teams, and healthcare stakeholders to troubleshoot infrastructure, networking, API, data pipeline, and deployment issues while maintaining architecture documentation, operational runbooks, and ETL process documentation using Git-based version control.

CLOUD ENGINEER

TECHNICAL SKILLS

Core Competencies:

- Azure Health Data Services (AHDS)
- Healthcare Interoperability
- FHIR R4 Standards
- SMART on FHIR
- CMS Interoperability Compliance
- Healthcare API Integration
- Cloud Architecture
- Infrastructure as Code (Terraform, Bicep)
- DevOps & CI/CD
- Security & Compliance

Azure Services & Technologies:

- Azure Health Data Services (FHIR Service)
- Azure API Management (APIM)
- Azure Functions
- Azure Key Vault
- Azure Monitor
- Log Analytics
- Application Insights
- Azure SQL Database
- Azure Storage Accounts
- Azure Application Gateway
- Azure Virtual Networks (VNet)
- Private Endpoints

Security & Identity:

- Microsoft Entra ID (Azure AD)
- OAuth 2.0
- SMART on FHIR Authorization
- RBAC
- Managed Identities
- HIPAA-Compliant Security Controls
- Azure Policy

DevOps & Automation:

- Azure DevOps
- Terraform
- Bicep
- Git
- ServiceNow Integration

Programming & Scripting:

- Python
- PowerShell
- Bash
- SQL

Cloud Engineer | Jul 2024 – Aug 2025

Designed and automated CI/CD pipelines using Azure DevOps and YAML for infrastructure provisioning and application deployment, implementing infrastructure automation using ARM/Bicep templates and enabling teams to provision resources through self-service Azure DevOps pipelines using managed identities, avoiding direct Azure permission assignment. Deployed and secured AKS clusters within a hub-spoke network, configuring private endpoints, NSGs, RBAC, and managed identity integration for containerized workloads. Configured Azure Virtual Desktop (AVD) environments from scratch including host pools, session hosts, and workspace setup, using GPO and Intune to push FSLogix profile settings uniformly across all session hosts at scale. Automated workflows using Azure Logic Apps, Function Apps, and Power Automate including ServiceNow ITSM integrations triggered by Azure Sentinel alerts. Managed Azure resources including Azure SQL, Synapse Analytics, and Data Lake with governance, tagging, and RBAC controls, while implementing monitoring using Azure Monitor and Log Analytics and optimizing log ingestion costs by filtering telemetry at source and setting retention policies aligned to compliance requirements.

DevOps Engineer | Sep 2020 – Jul 2022

Supported the deployment and management of Azure cloud infrastructure, including Virtual Machines, Virtual Networks, Storage Accounts, Azure SQL Database, and Key Vault. Assisted in building and maintaining CI/CD pipelines using Azure DevOps, Git, and YAML to automate application and infrastructure deployments across development and test environments. Developed PowerShell and Python scripts to automate routine operational tasks, resource provisioning, and environment configuration. Implemented monitoring and alerting solutions using Azure Monitor, Log Analytics, and Application Insights to improve platform visibility and incident response. Worked with Azure Data Factory and SQL-based data pipelines to support data integration, transformation, and reporting requirements. Assisted in implementing security controls including RBAC, Network Security Groups, and Key Vault-based secret management. Collaborated with development and infrastructure teams on troubleshooting, performance optimization, environment support, and technical documentation activities.



EDUCATION:

University of Massachusetts, Lowell, USA

2022 - 2024

Master of Science in Information Technology

Bharath University, India

2018 - 2022

Bachelor of Engineering